

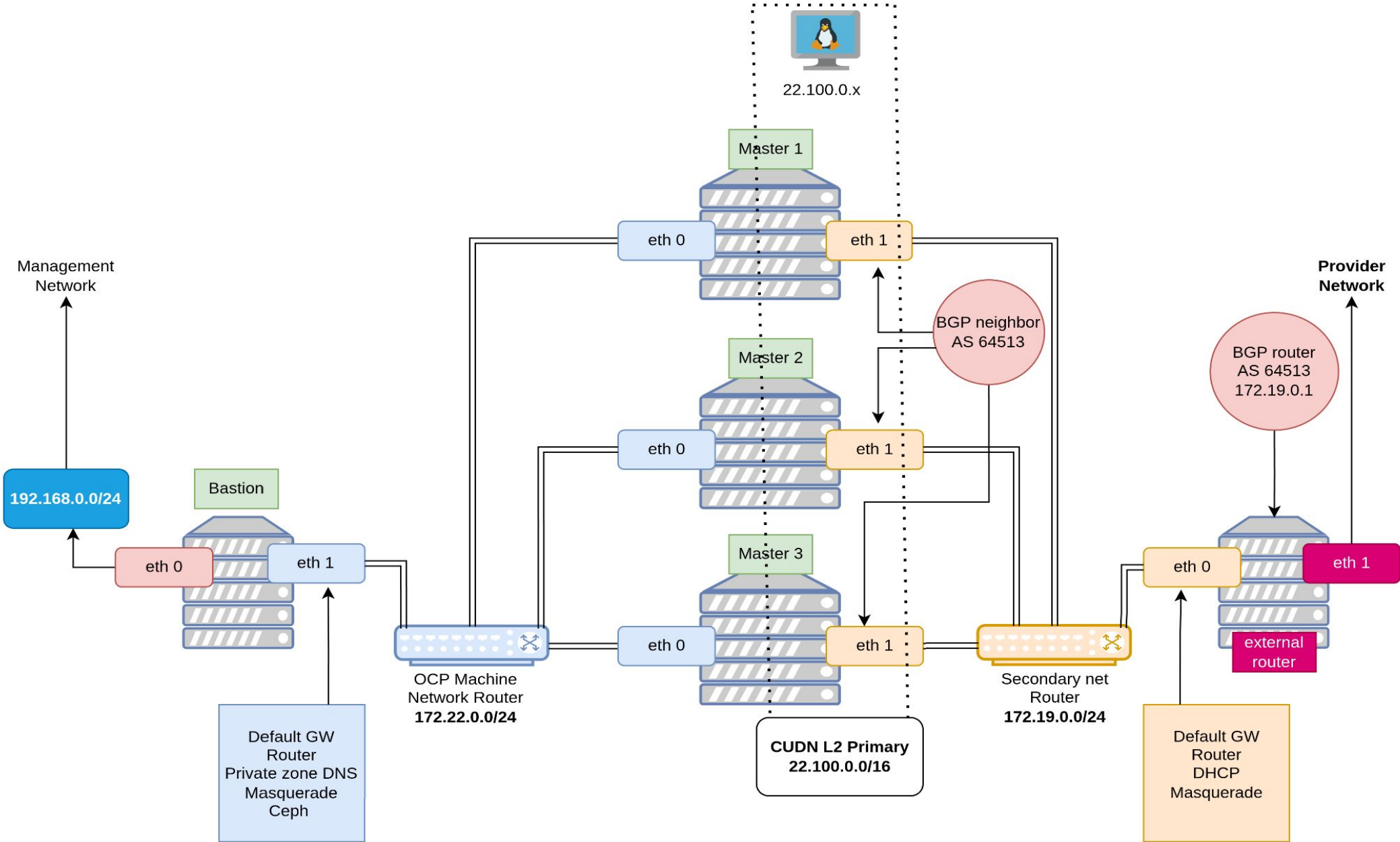
Virtualization Hackathon 2026 - Brno

OpenShift Virtualization BGP+EVPN

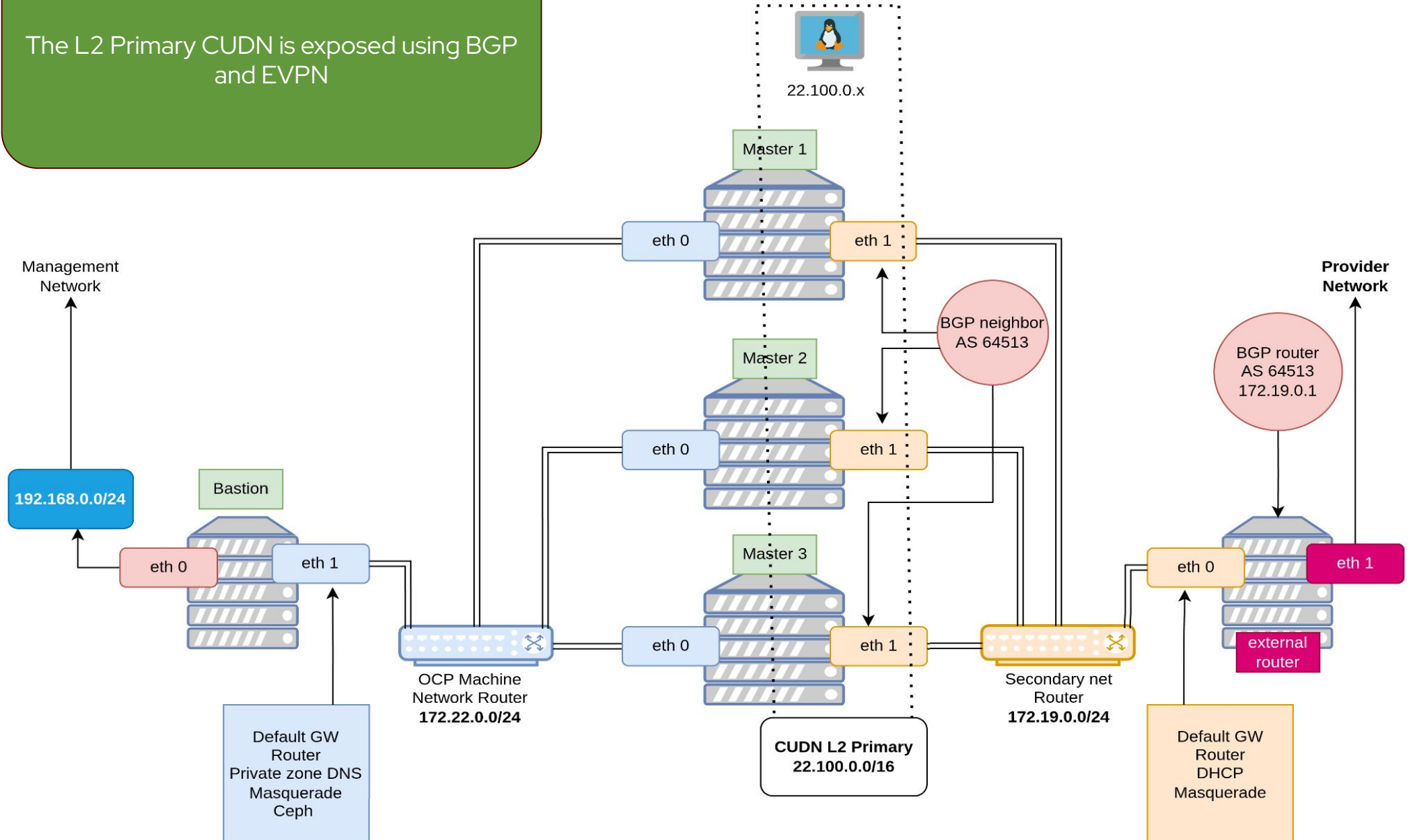
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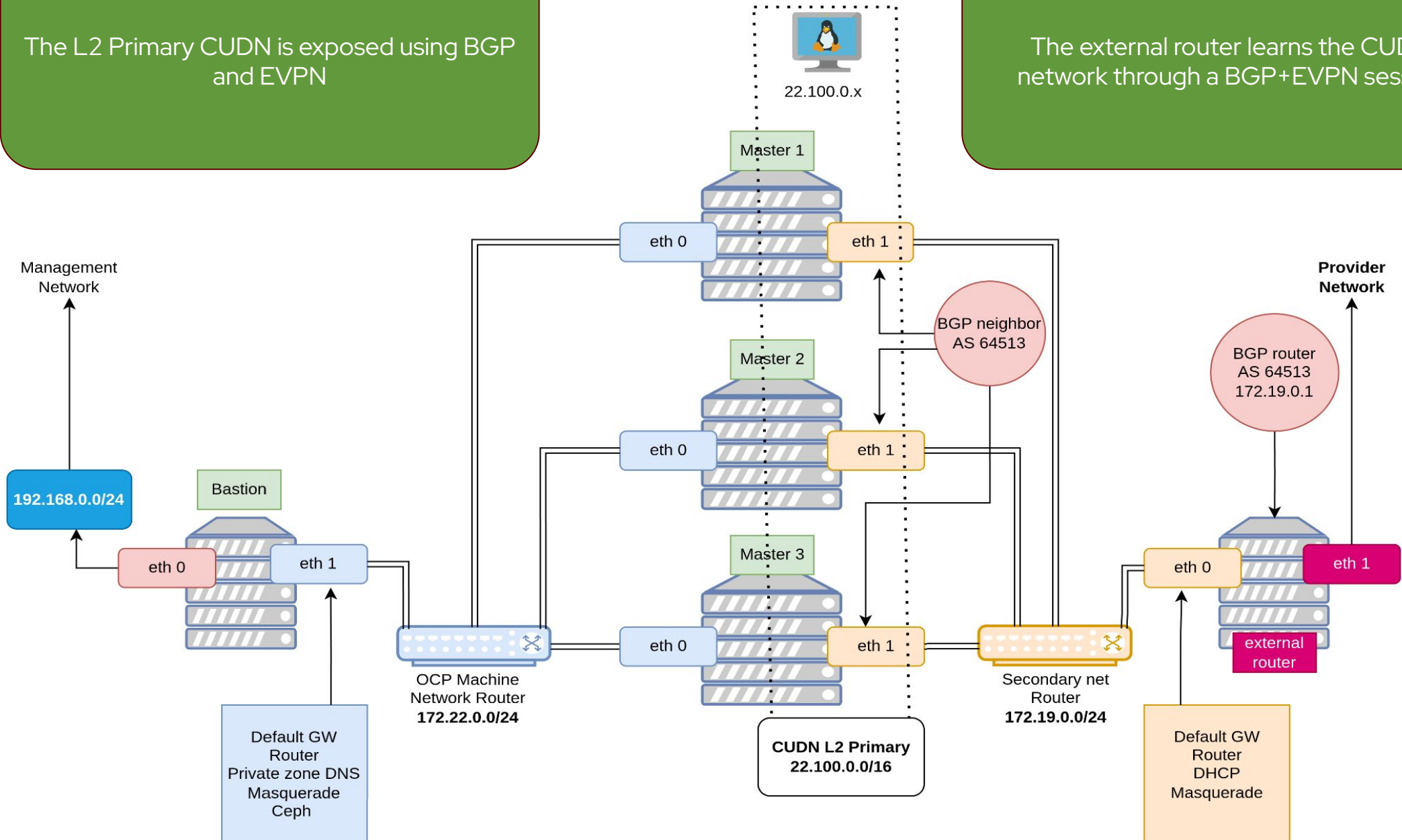


The L2 Primary CUDN is exposed using BGP and EVPN



The L2 Primary CUDN is exposed using BGP and EVPN

The external router learns the CUDN network through a BGP+EVPN session



Steps to be performed

OpenShift 4.22.0	External Router
Enable BGP	Create a bridge interface
Set static IPV4 for the secondary physical NICs	Create a vxlan interface
Create the VTEP	Create the FRR router configuration
Create the FRR Configuration	Check for imported VM's route
Create the Namespace	
Create the CUDN	
Create the Route Advertisement	
Create the VM	

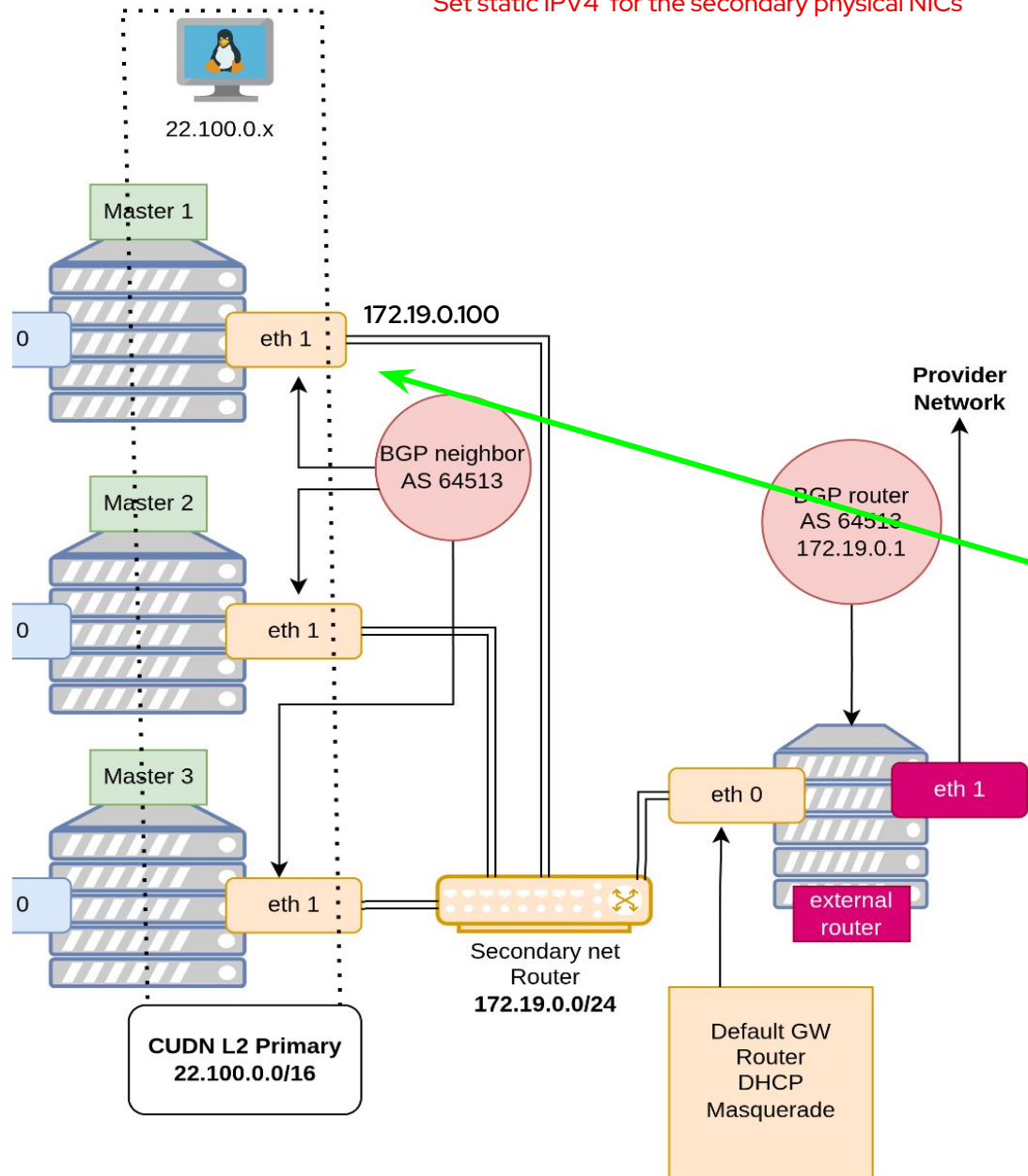


```
apiVersion: operator.openshift.io/v1
kind: Network
metadata:
  name: cluster
spec:
  additionalRoutingCapabilities:
    providers:
    - FRR
  ...
  defaultNetwork:
    ovnKubernetesConfig:
      gatewayConfig:
        ipForwarding: "Global"
        routingViaHost: true
        routeAdvertisements: Enabled
  type: OVNKubernetes
```

One time operation



Set static IPV4 for the secondary physical NICs



apiVersion: nmstate.io/v1

kind: NodeNetworkConfigurationPolicy

metadata:

name: node-1-fixed-ip

spec:

nodeSelector:

node: "1"

desiredState:

interfaces:

- **name:** eth1

type: ethernet

state: up

ipv4:

dhcp: false

address:

- **ip:** 172.19.0.100

prefix-length: 24

enabled: true

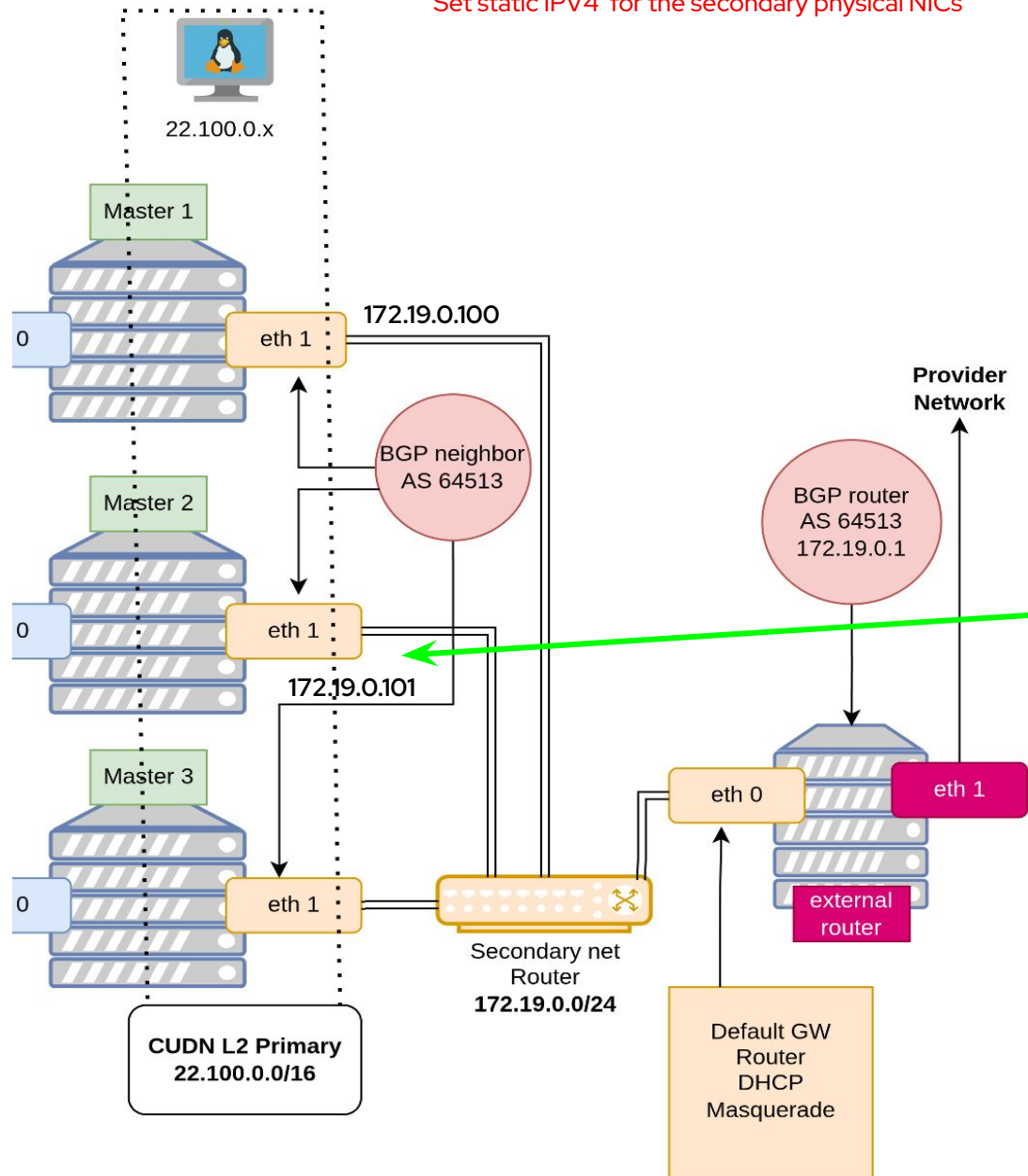
ipv6:

enabled: false

Repeat for each node
3 nodes = 3 NNCP
Each node has his own
label (node: "1", node:
"2" and node: "3")



Set static IPV4 for the secondary physical NICs



apiVersion: nmstate.io/v1

kind: NodeNetworkConfigurationPolicy

metadata:

name: node-2-fixed-ip

spec:

nodeSelector:

node: "2"

desiredState:

interfaces:

- name: eth1

type: ethernet

state: up

ipv4:

dhcp: false

address:

- ip: 172.19.0.101

prefix-length: 24

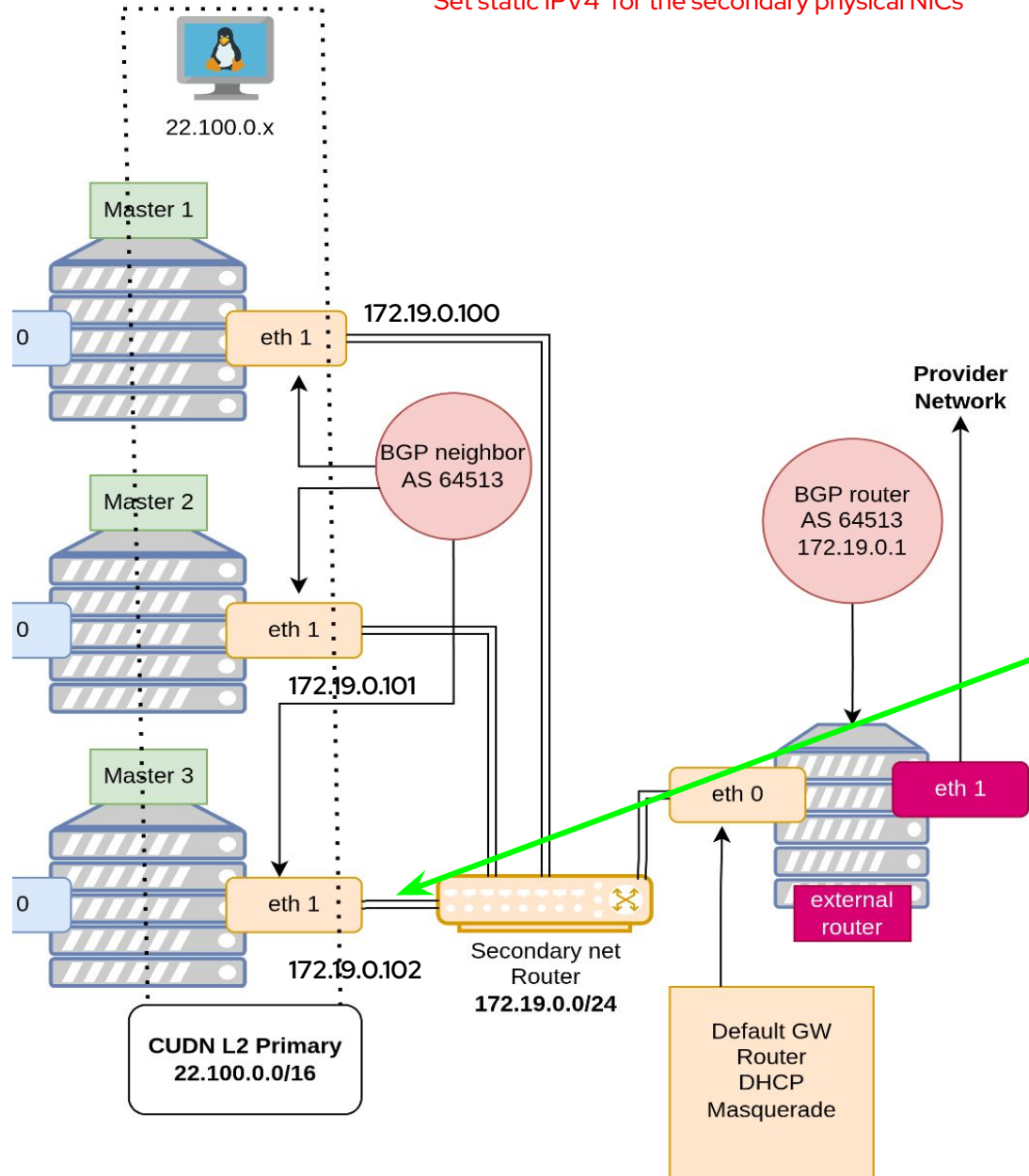
enabled: true

ipv6:

enabled: false



Set static IPV4 for the secondary physical NICs



apiVersion: nmstate.io/v1

kind: NodeNetworkConfigurationPolicy

metadata:

name: node-3-fixed-ip

spec:

nodeSelector:

node: "3"

desiredState:

interfaces:

- name: eth1

type: ethernet

state: up

ipv4:

dhcp: false

address:

- ip: 172.19.0.102

prefix-length: 24

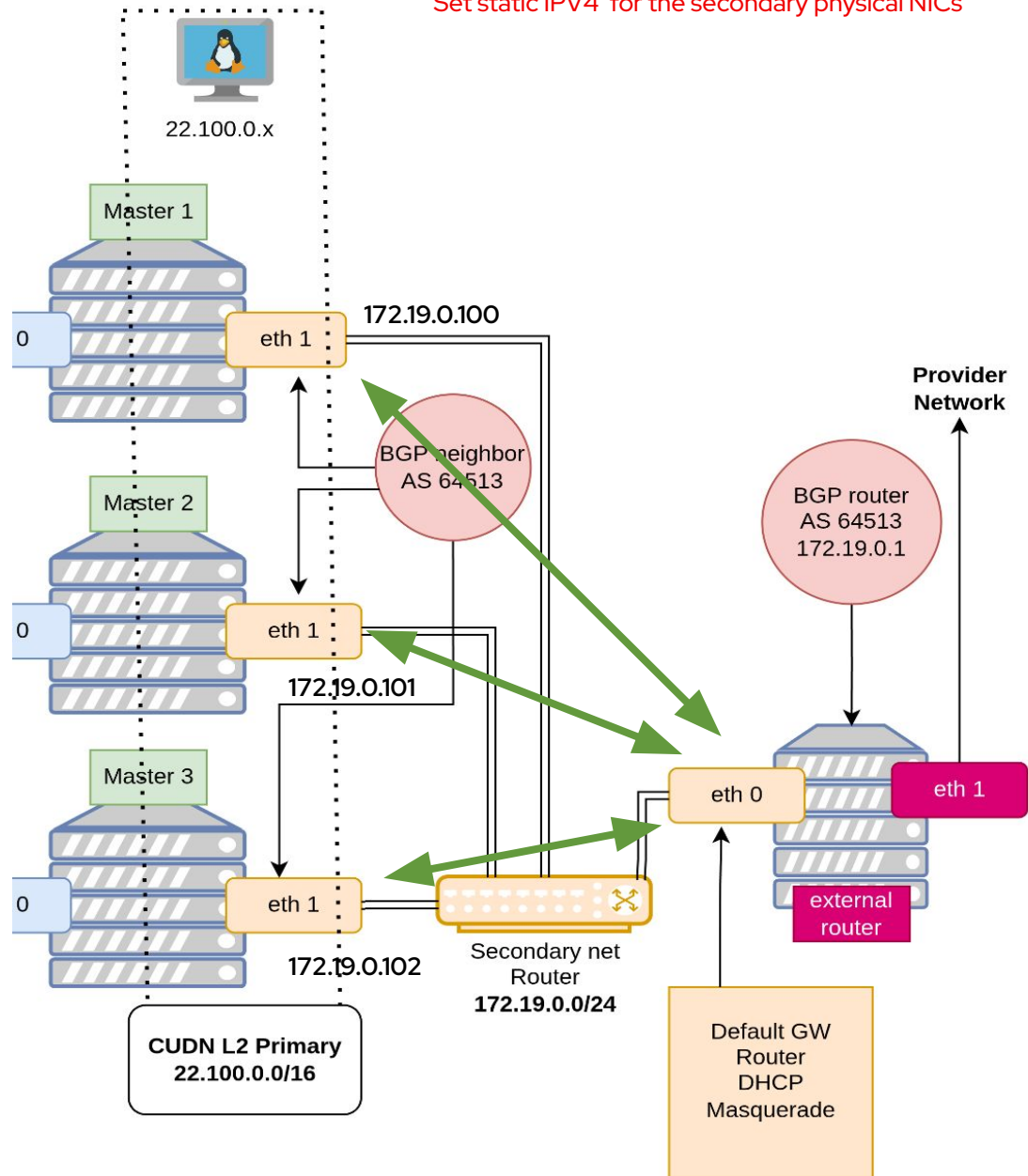
enabled: true

ipv6:

enabled: false



Set static IPV4 for the secondary physical NICs



Check connectivity from the external router to the OCP nodes secondary physical nic (eth1)

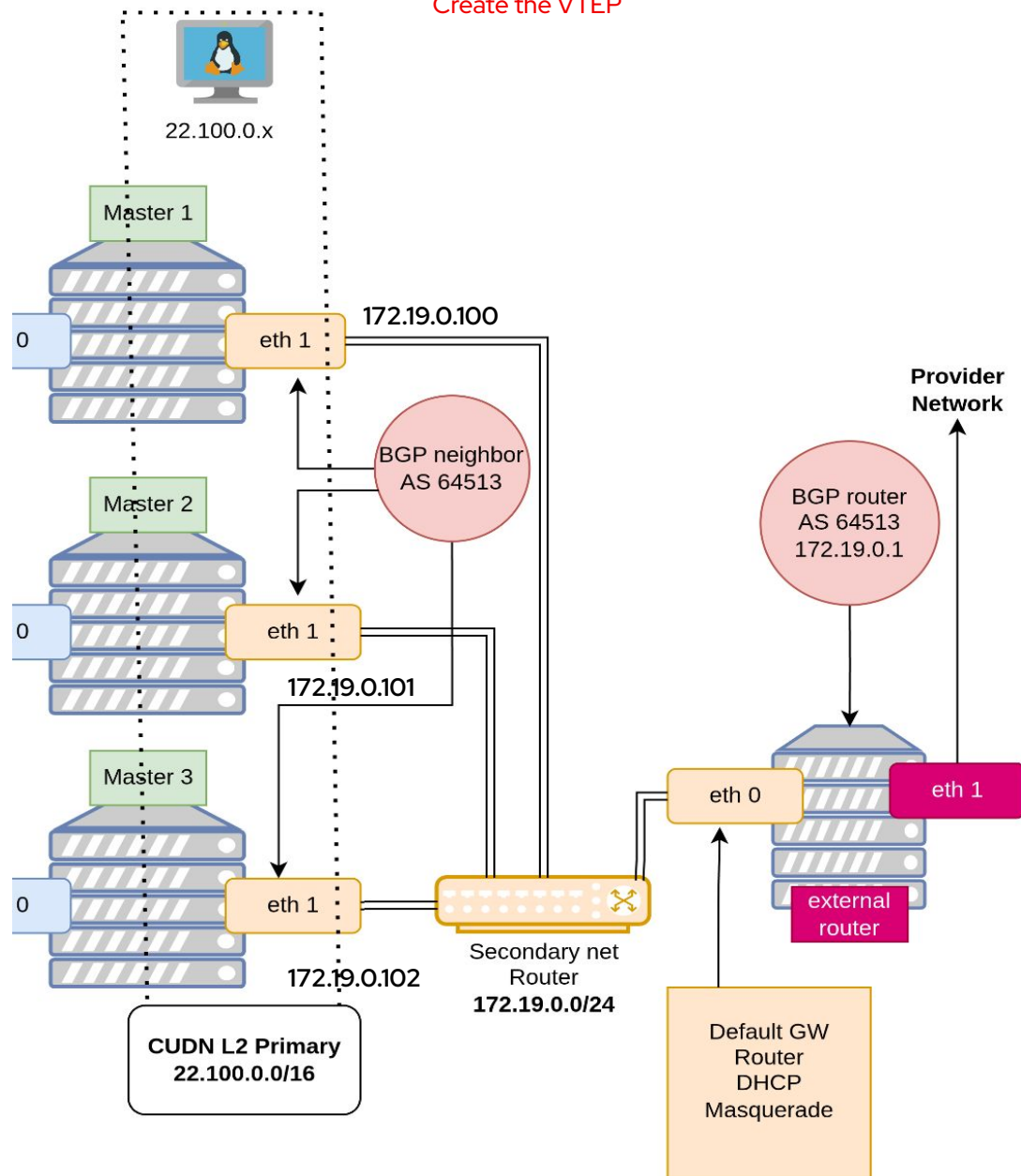
```
vale@external-router:~$ for h in 172.19.0.100 172.19.0.101 172.19.0.102; do ping -c 3 $h & done; wait
```

```
64 bytes from 172.19.0.100: icmp_seq=1 ttl=64 time=0.238 ms  
64 bytes from 172.19.0.101: icmp_seq=1 ttl=64 time=0.261 ms  
64 bytes from 172.19.0.102: icmp_seq=1 ttl=64 time=0.637 ms
```

BGP is not involved yet



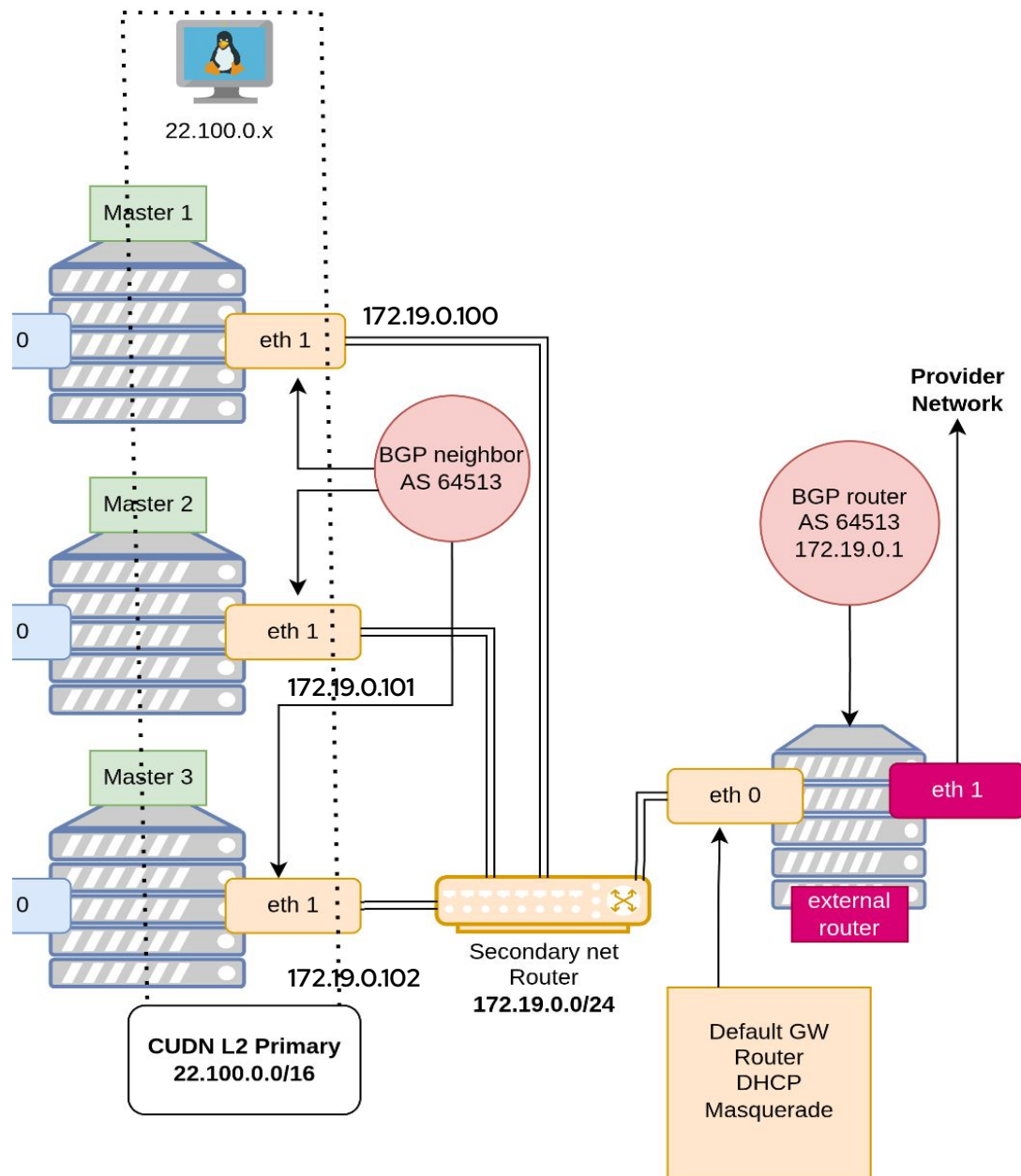
Create the VTEP



```
apiVersion: k8s.ovn.org/v1
kind: VTEP
metadata:
  name: evpn-vtep
spec:
  cidrs:
    - 172.19.0.0/24
  mode: Unmanaged
```

- The IPV4s assigned to the secondary physical NICs, must be part of the VTEP's cidrs
- As for OCP 4.22.0, only the "Unmanaged" mode is available
- The VTEP controller checks for the node ["k8s.ovn.org/host-cidrs"](https://k8s.ovn.org/host-cidrs) annotation
- If the VTEP controller finds a node with an IPV4 being part of the VTEP's specified cidrs, it will add a new ["k8s.ovn.org/vteps"](https://k8s.ovn.org/vteps) annotation to the node, containing the vtep's assigned IPV4
- The VTEP will use the node IPV4 assigned before to the secondary physical NIC

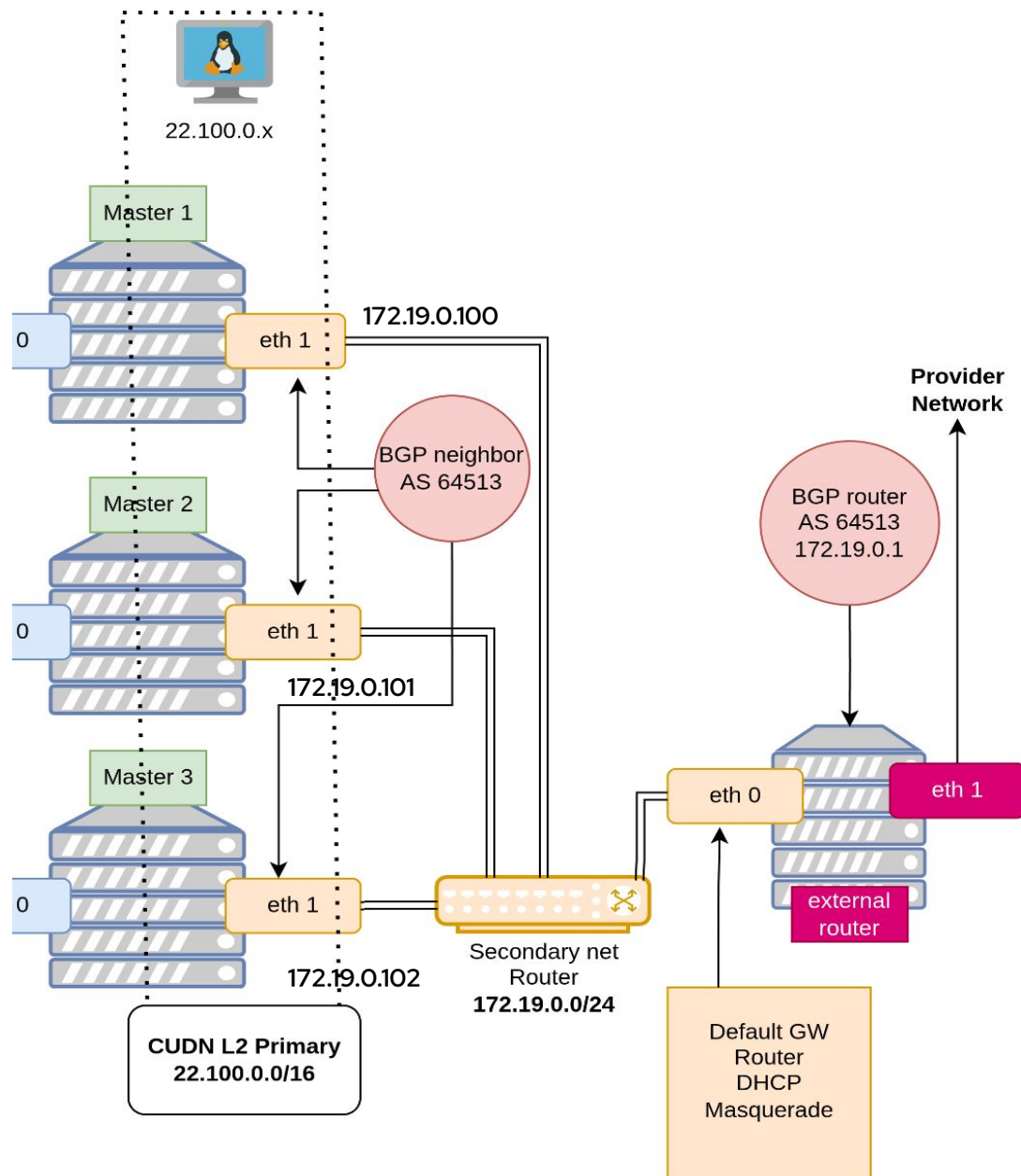




```
~/Projects/deolab$ oc get nodes -o yaml | grep cidrs
k8s.ovn.org/host-cidrs: ["172.19.0.100/24","172.22.0.11/24"]
k8s.ovn.org/host-cidrs: ["172.19.0.101/24","172.22.0.12/24"]
k8s.ovn.org/host-cidrs: ["172.19.0.102/24","172.22.0.13/24"]
```

- The 172.22.0.0/24 is the OCP' Machine Network
- The 172.19.0.0/24 is the Secondary Network (**the one we want to use for the VTEP**)





```
~/Projects/deolab$ oc get vtep
NAME          ACCEPTED REASON
evpn-vtep    True      Allocated
```

- Because the VTEP is accepted, a new vtep annotation ["ik8s.ovn.org/vteps"](https://ik8s.ovn.org/vteps) is added to the OCP nodes

```
oc get nodes -o yaml | grep vteps
k8s.ovn.org/vteps: '{"evpn-vtep":{"ips":["172.19.0.100"]}}'
k8s.ovn.org/vteps: '{"evpn-vtep":{"ips":["172.19.0.101"]}}'
k8s.ovn.org/vteps: '{"evpn-vtep":{"ips":["172.19.0.102"]}}'
```



The diagram illustrates a network topology for a multi-master CUDN L2 Primary setup. On the left, three server racks labeled 'Master 1', 'Master 2', and 'Master 3' are connected to a 'CUDN L2 Primary' network with IP range 22.100.0.0/16. Each master has an 'eth 1' interface. These interfaces are connected to a 'Secondary net Router' (172.19.0.0/24) which has interfaces 'eth 1' (172.19.0.100), 'eth 1' (172.19.0.101), and 'eth 1' (172.19.0.102). The Secondary net Router is also connected to a 'BGP neighbor AS 64513' (172.19.0.100). The BGP neighbor AS is connected to a 'BGP router AS 64513' (172.19.0.1) which is connected to an 'external router' (eth 1). The external router is connected to a 'Default GW Router DHCP Masquerade' (eth 0) which is connected to the 'Provider Network'.

```
apiVersion: frrk8s.metallb.io/v1beta1
kind: FRRConfiguration
metadata:
  name: underlay-peering
  namespace: openshift-frr-k8s
  labels:
    evpn: "true"
spec:
  nodeSelector: {}
  bgp:
    routers:
      - asn: 64513
    neighbors:
      - address: 172.19.0.1
        asn: 64513
        disableMP: true
        dualStackAddressFamily: false
        toAdvertise:
          allowed:
            mode: all
        toReceive:
          allowed:
            mode: all
```



The diagram illustrates a network topology for a multi-master CUDN L2 Primary setup. On the left, three server racks labeled "Master 1", "Master 2", and "Master 3" are connected to a "CUDN L2 Primary" network with IP range 22.100.0.0/16. Each master has an "eth 1" interface. These interfaces are connected to a "Secondary net Router" (172.19.0.0/24) which has interfaces "eth 1" (172.19.0.100), "eth 1" (172.19.0.101), and "eth 1" (172.19.0.102). The router is also connected to a "BGP neighbor AS 64513". To the right, a "Default GW Router" (DHCP, Masquerade) is connected to the "Secondary net Router" via "eth 0". This default gateway is also connected to an "external router" via "eth 1". The external router is connected to a "BGP router AS 64513" (172.19.0.1), which is finally connected to the "Provider Network".

NAME _____ AGE _____

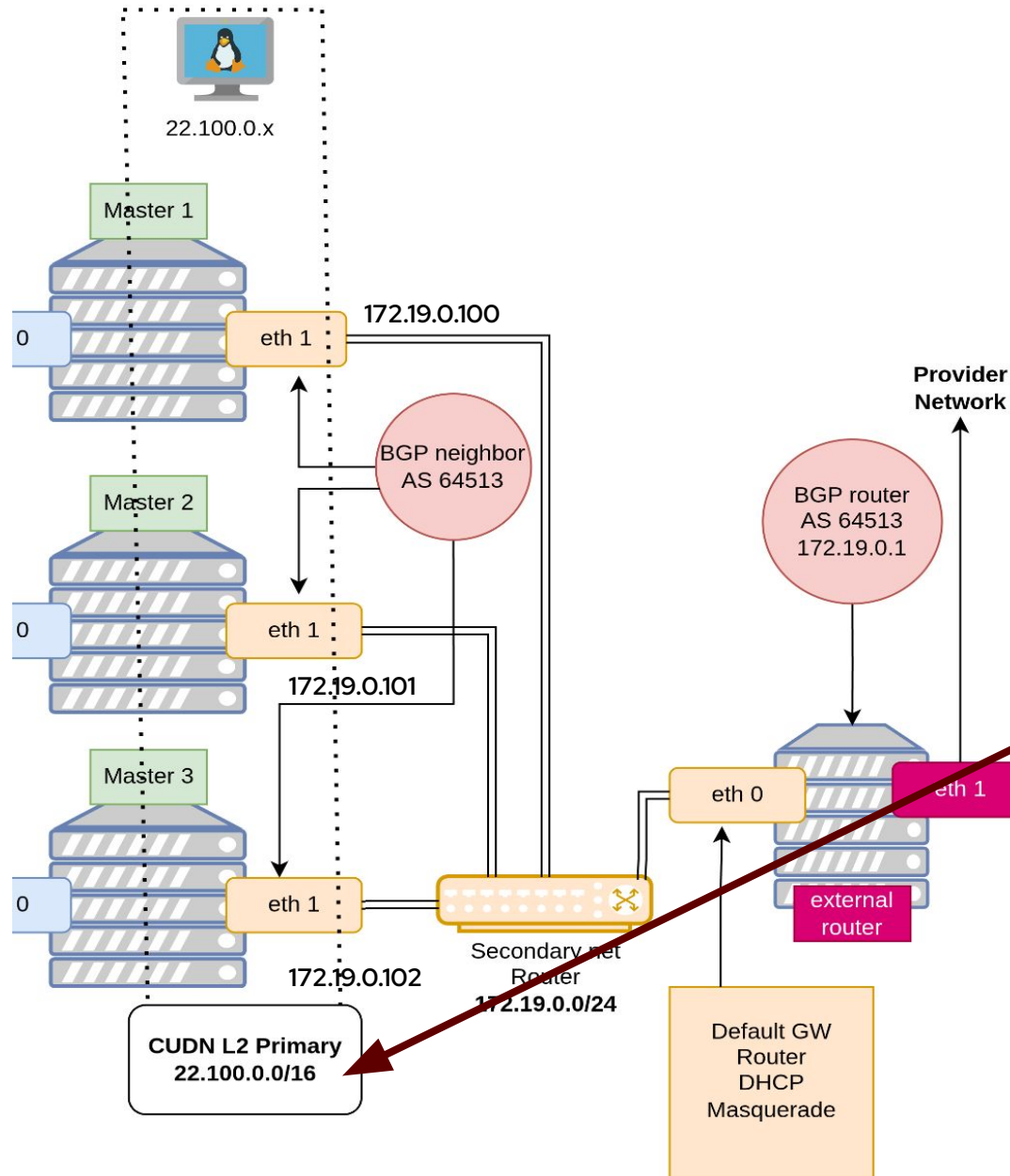
ovnk-generated-clz95 40h

underlay-peering 40h

- If the “underlay-peering” FRRConfiguration is accepted, new FRRConfigurations will be created for each node. In this lab there are 3 nodes
- **The FRRConfiguration also features the “nodeSelector” functionality.**



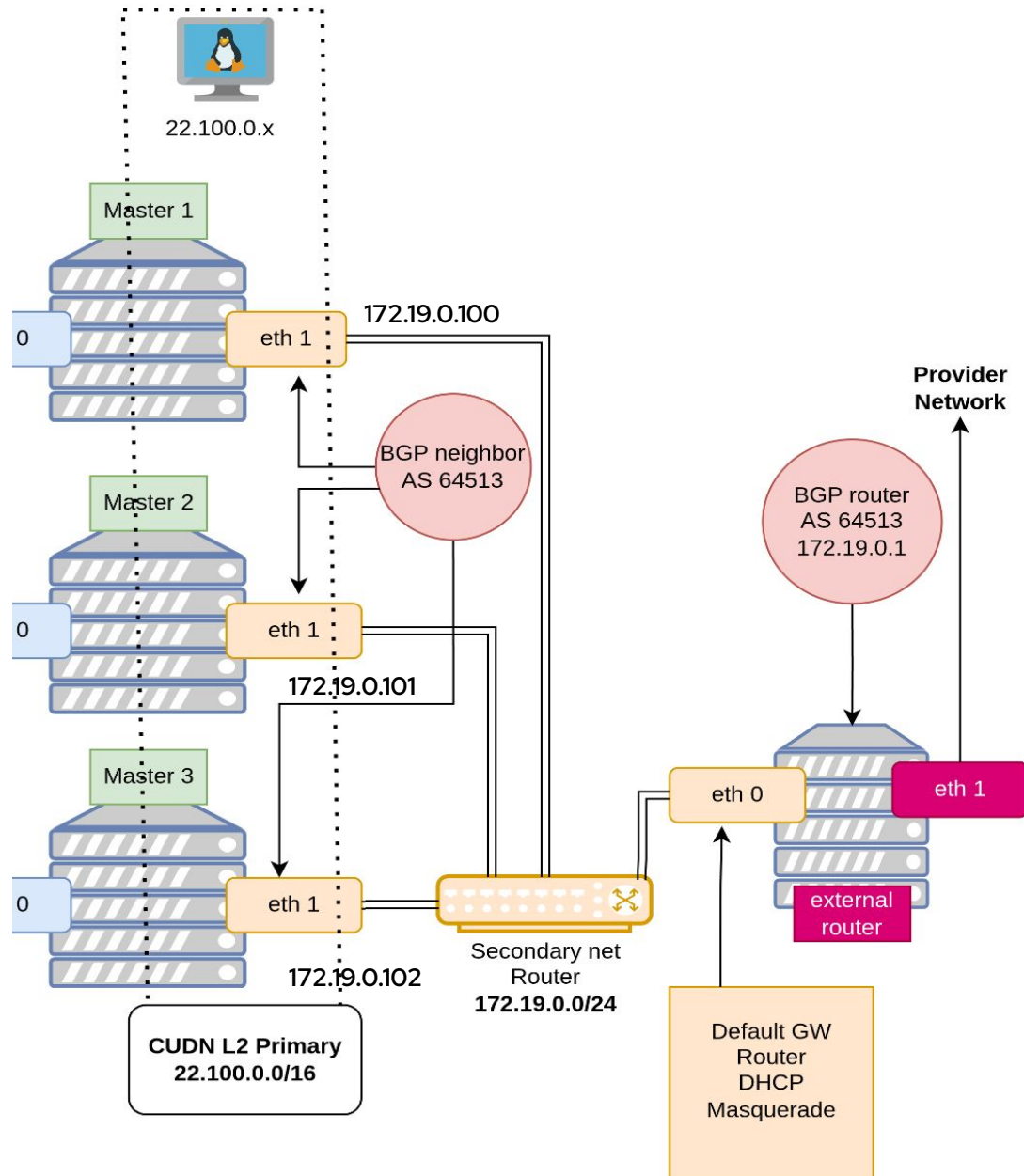
Create the CUDN



```
apiVersion: k8s.ovn.org/v1
kind: ClusterUserDefinedNetwork
metadata:
  labels:
    advertise: "true"
    name: finance
spec:
  namespaceSelector:
    matchLabels:
      network: finance
  network:
    layer2:
      role: Primary
      subnets:
        - 22.100.0.0/16
      topology: Layer2
      transport: EVPN
    evpn:
      vtep: evpn-vtep
      macVRF:
      vni: 100
```



Create the CUDN

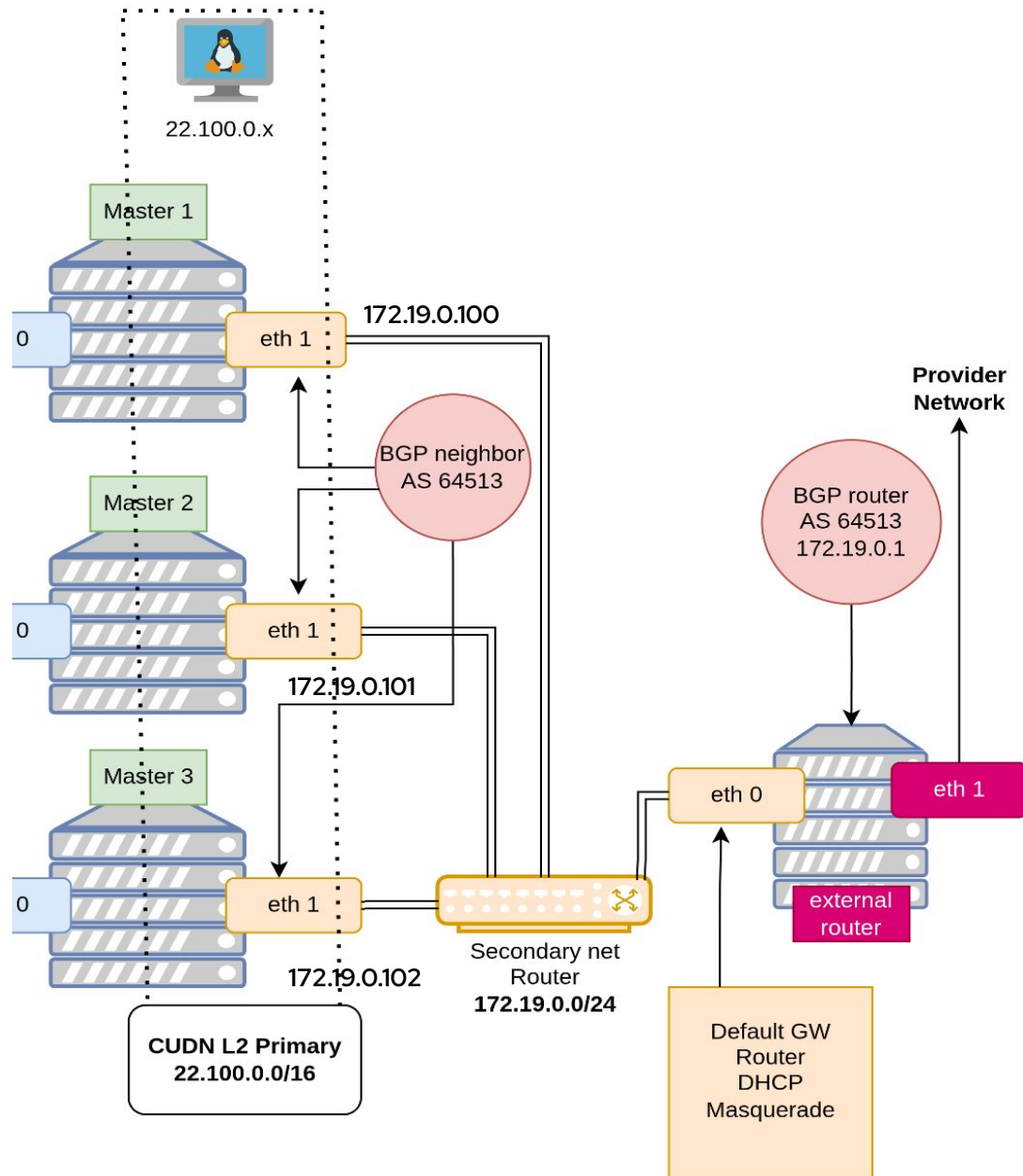


```
apiVersion: k8s.ovn.org/v1
kind: ClusterUserDefinedNetwork
metadata:
  labels:
    advertise: "true"
    name: finance
spec:
  namespaceSelector:
    matchLabels:
      network: finance
  network:
    layer2:
      role: Primary
      subnets:
        - 22.100.0.0/16
    topology: Layer2
  transport: EVPN
  evpn:
    vtep: evpn-vtep
    macVRF:
      vni: 100
```

- transport: EVPN



Create the CUDN

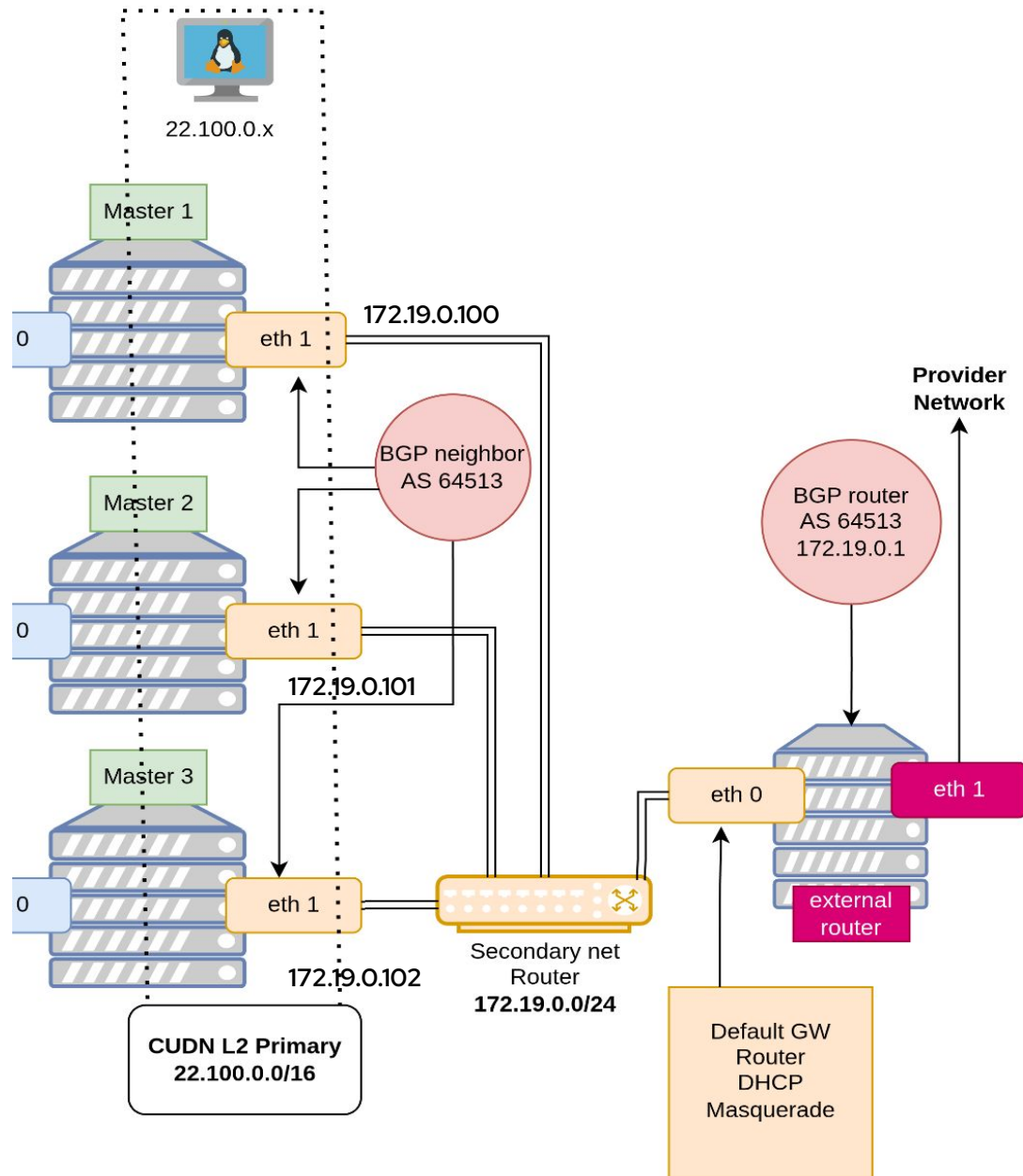


```
apiVersion: k8s.ovn.org/v1
kind: ClusterUserDefinedNetwork
metadata:
  labels:
    advertise: "true"
    name: finance
spec:
  namespaceSelector:
    matchLabels:
      network: finance
  network:
    layer2:
      role: Primary
      subnets:
        - 22.100.0.0/16
    topology: Layer2
    transport: EVPN
  evpn:
    vtep: evpn-vtep
    macVRF:
      vni: 100
```

- Targeting the before created VTEP



Create the CUDN

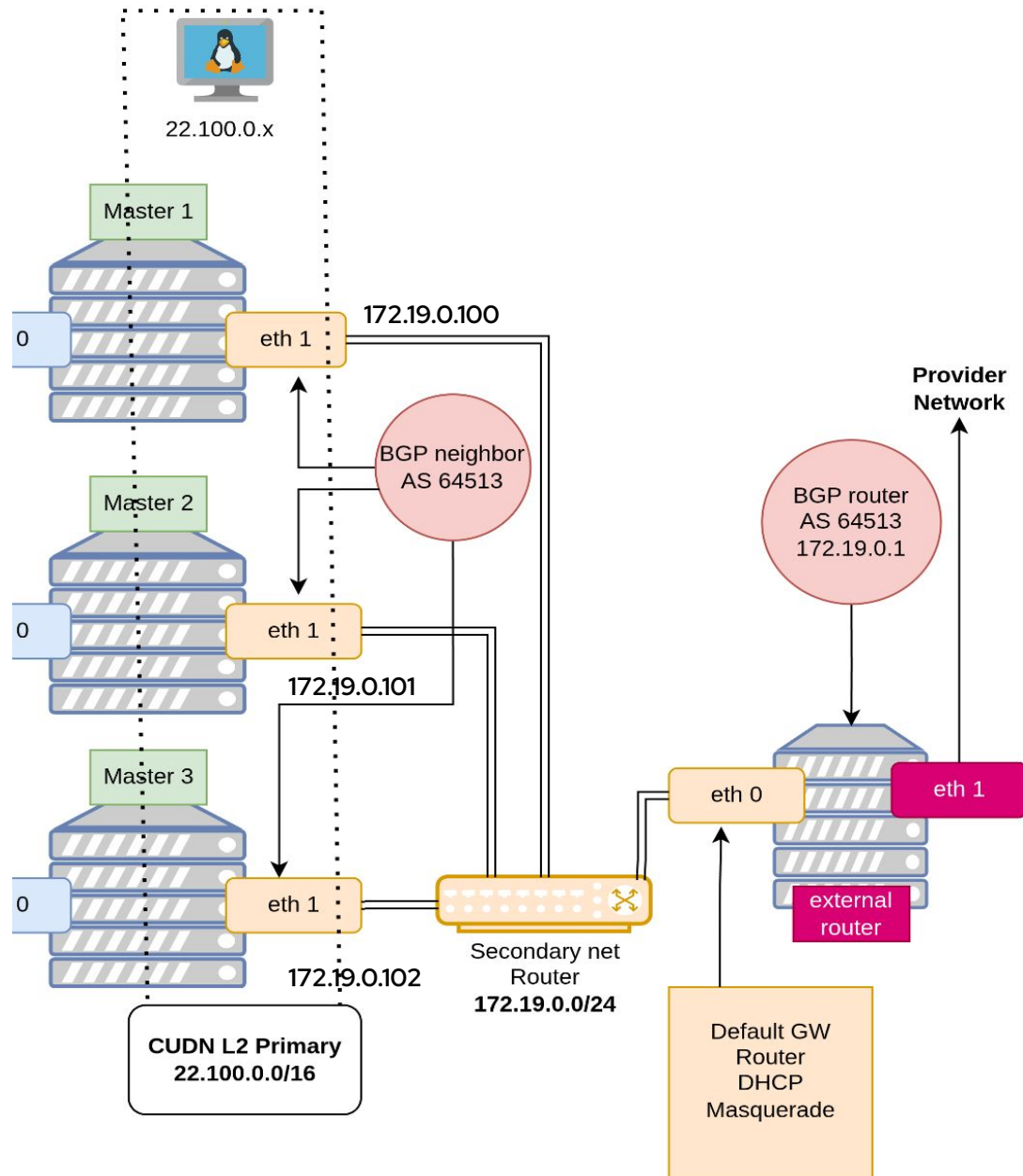


```
apiVersion: k8s.ovn.org/v1
kind: ClusterUserDefinedNetwork
metadata:
  labels:
    advertise: "true"
    name: finance
spec:
  namespaceSelector:
    matchLabels:
      network: finance
  network:
    layer2:
      role: Primary
      subnets:
        - 22.100.0.0/16
    topology: Layer2
    transport: EVPN
  evpn:
    vtep: evpn-vtep
  macVRF:
    vni: 100
```

- Use a MAC-VRF with a VXLAN VNI=100 (L2VNI)



Create the Route Advertisement

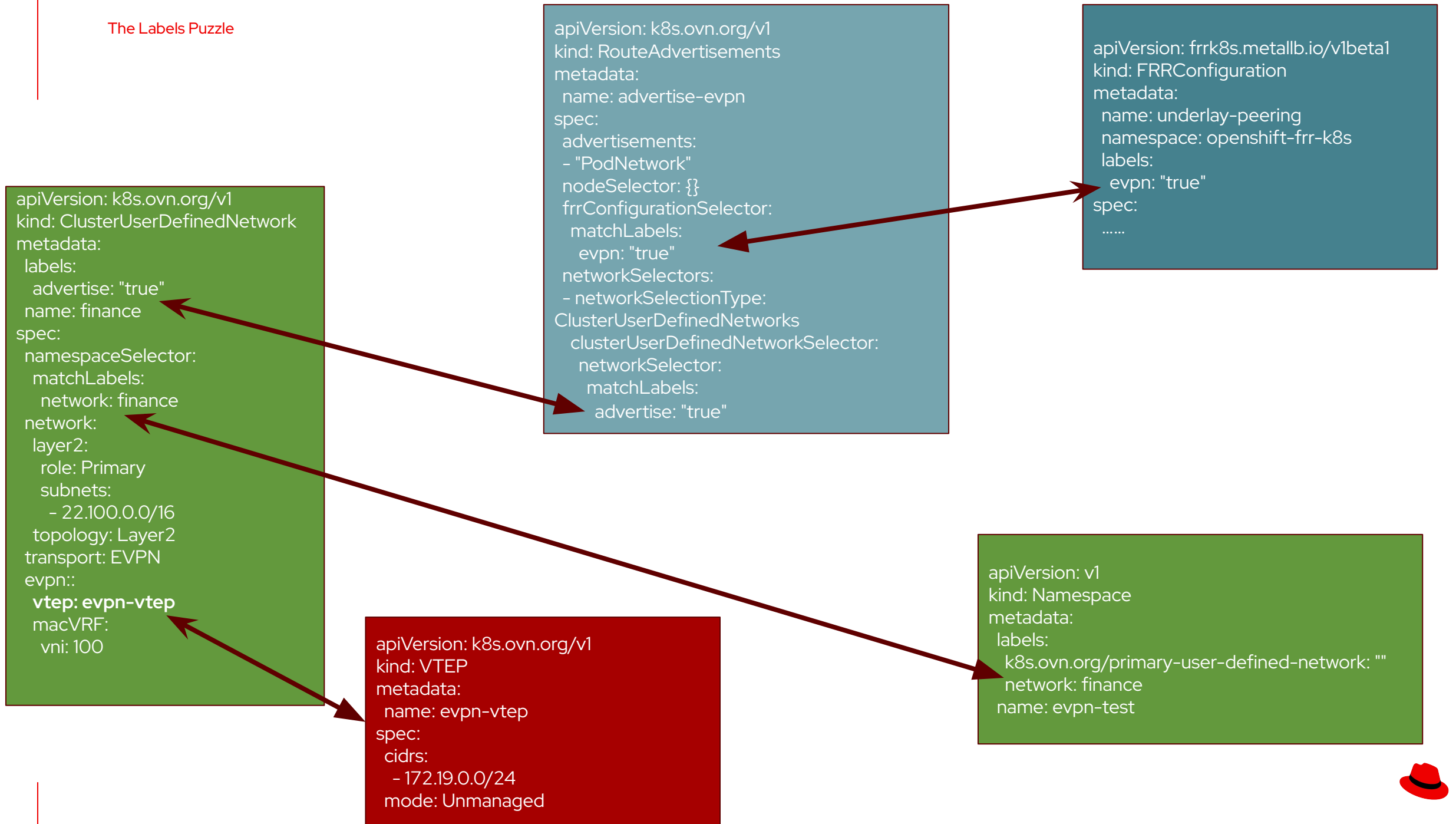


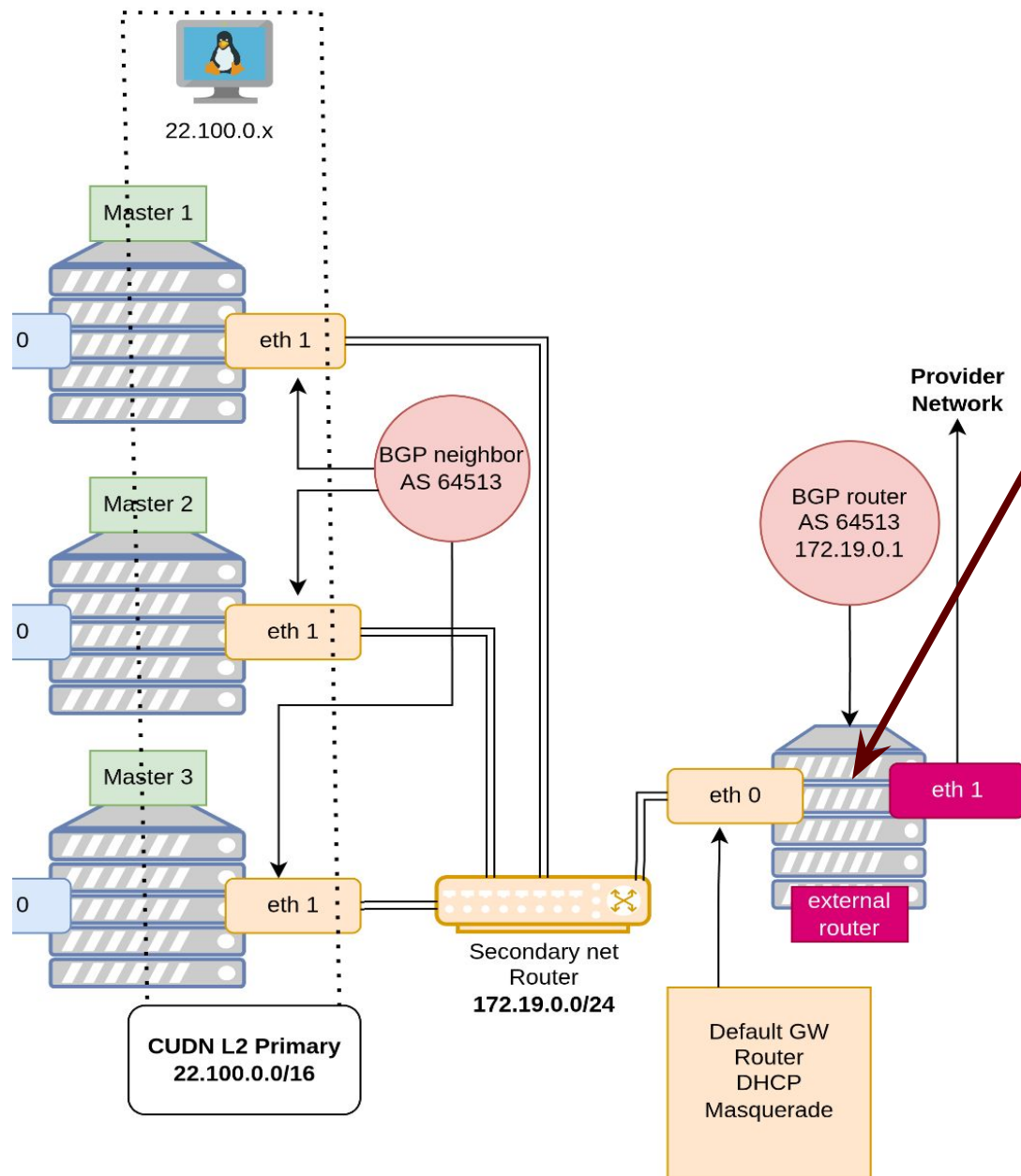
```
apiVersion: k8s.ovn.org/v1
kind: RouteAdvertisements
metadata:
  name: advertise-evpn
spec:
  advertisements:
    - "PodNetwork"
  nodeSelector: {}
  frrConfigurationSelector:
    matchLabels:
      evpn: "true"
  networkSelectors:
    - networkSelectionType:
        ClusterUserDefinedNetworks
        clusterUserDefinedNetworkSelector:
          networkSelector:
            matchLabels:
              advertise: "true"
```

- Pay attention to all the "Selectors"



The Labels Puzzle





External Router Configuration

- Configuring an external BGP/EVPN router should not be needed during demoing the OCP BGP/EVPN capabilities on the customer site, as we should expect BGP/EVPN/VXLAN switches already in place.
- The external router used here, it is a standard Fedora 44 linux box, running the FRR BGP router as a container



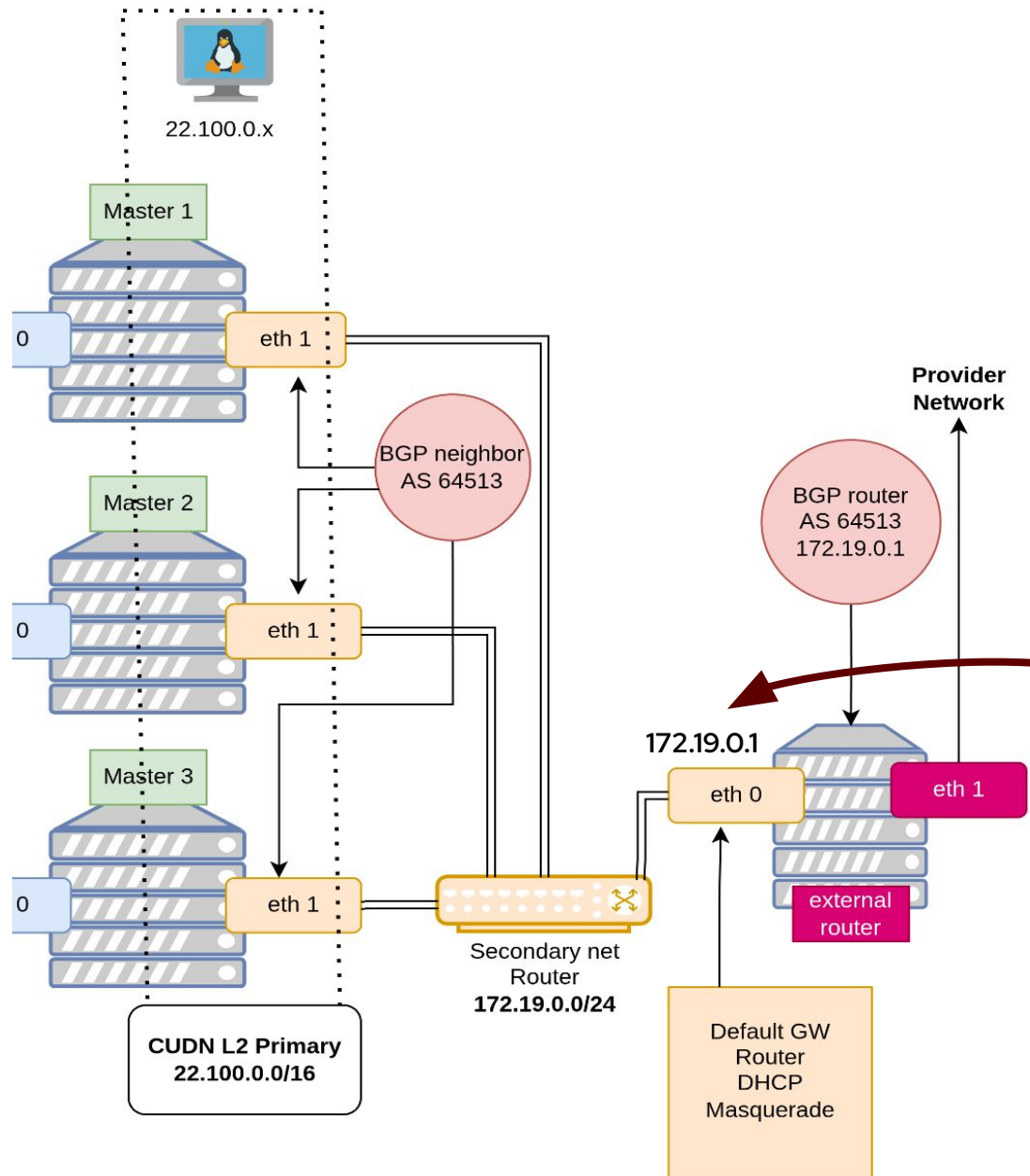
External Router Configuration

```
apiVersion: k8s.ovn.org/v1
kind: ClusterUserDefinedNetwork
metadata:
  labels:
    advertise: "true"
  name: finance
spec:
  namespaceSelector:
    matchLabels:
      network: finance
  network:
    layer2:
      role: Primary
      subnets:
        - 22.100.0.0/16
    topology: Layer2
  transport: EVPN
  evpn:
    vtep: evpn-vtep
    macVRF:
      vni: 100
```

- The VXLAN ID (VNI) configured for the CUDN is 100
- The steps to configure the external router are:
 - Create a linux bridge
 - Create the VXLAN with the id=100
 - Attach the VXLAN to the linux bridge
 - Assign an IPV4 address to the bridge
 - The Bridge's IPv4 address must be part of the subnet configured in the CUDN



External Router Configuration



Linux bridge

```
ip link add br100 type bridge  
ip link set br100 up
```

VXLAN setup

```
ip link add vxlan100 type vxlan id 100 dstport 4789 local 172.19.0.1 nolearning
```

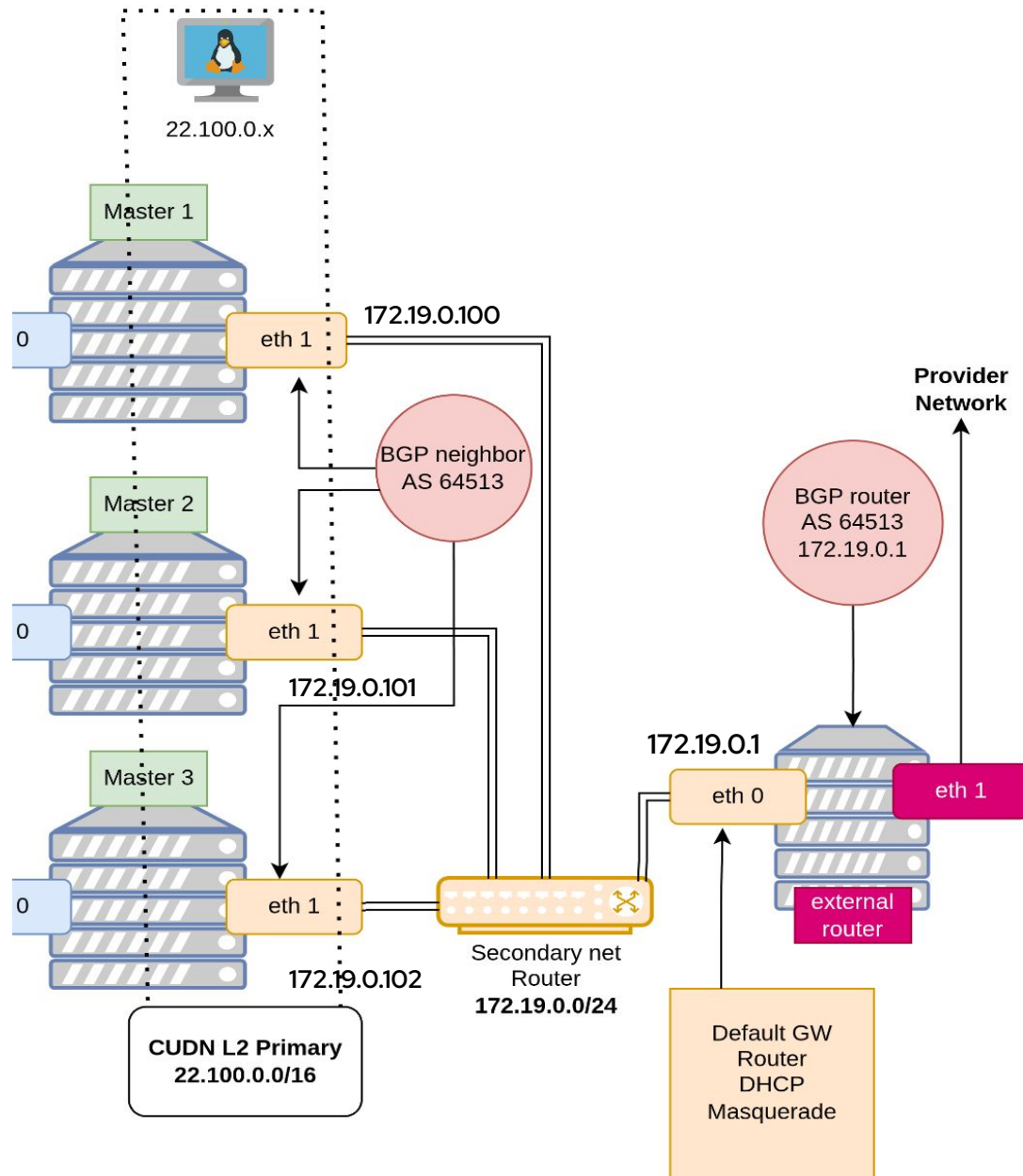
```
ip link set vxlan100 master br100  
ip link set vxlan100 up
```

Bridge IPv4

```
ip addr add 22.100.0.1/16 dev br100
```

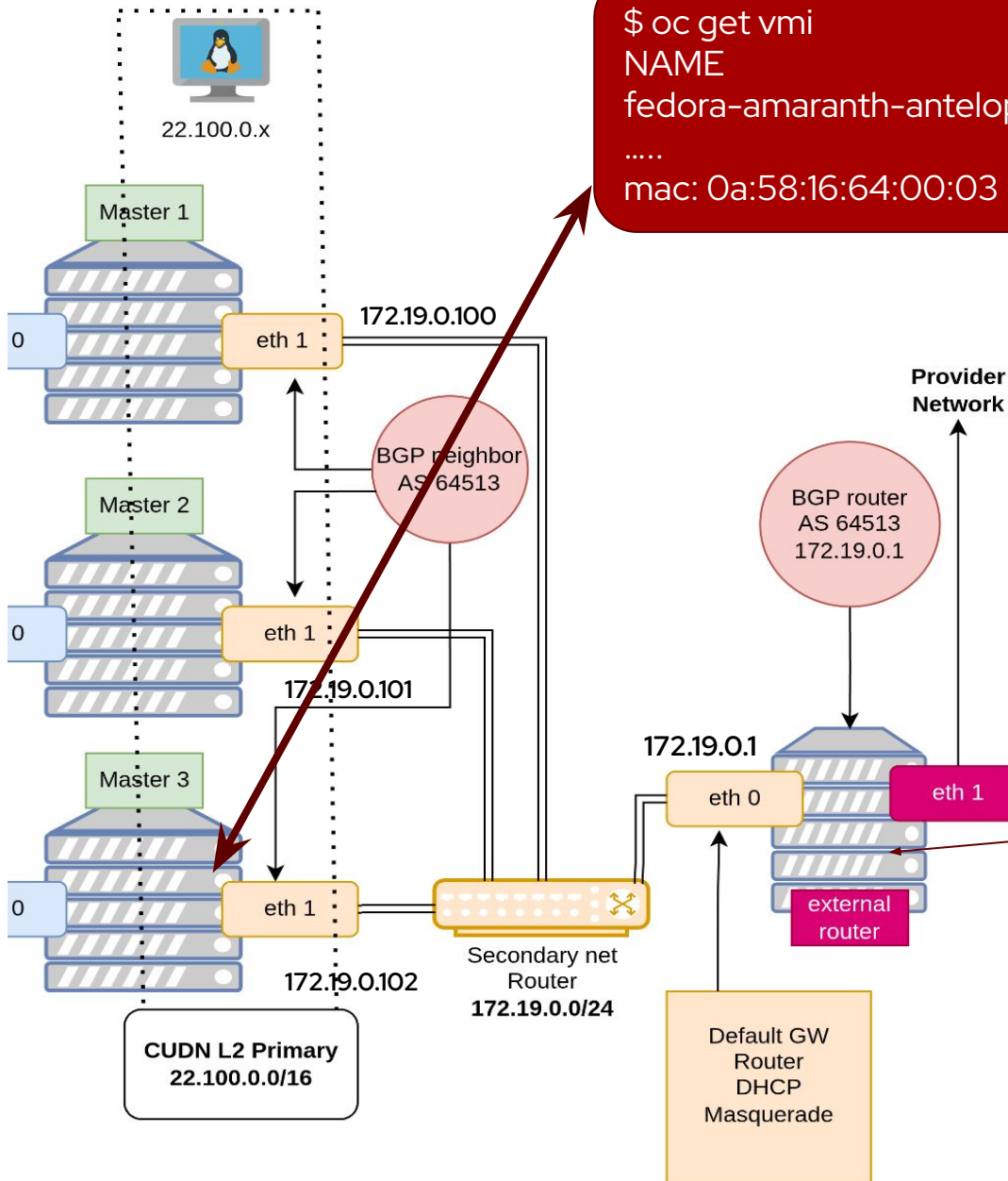


External Router Configuration



- Start the FRR BGP Router and check for imported EVPN mac Addresses
- We should see the VM's MAC, IPv4 and the assigned thanks to the BGP/EVPN/VXLAN communication
- The route should target the VTEP IP of the node where the VM is running
 - 172.19.0.100 (node 1)
 - 172.19.0.101 (node 2)
 - 172.19.0.102 (node 3)





```
$ oc get vmi
NAME                                AGE  PHASE  IP            NODENAME
fedora-amaranth-antelop2-2        59m  Running 22.100.0.3    58-47-ca-7e-51-e4
....
mac: 0a:58:16:64:00:03
```

FRR Router command -> "show evpn mac vni 100"

Number of MACs (local and remote) known for this VNI: 2

Flags: N=sync-neighs, I=local-inactive, P=peer-active, X=peer-proxy

MAC	Type	Flags	Intf/Remote	ES/VTEP	VLAN	Seq #'s
0a:58:16:64:00:03	remote		172.19.0.102			1/0
b6:bb:1c:86:82:77	local		br100		1	0/0

VM's MAC received
172.19.0.102 is the VTEP of the
OCP node where the VM is running



Migrate VirtualMachine to a different Node

Select the target Node to migrate your VirtualMachine to.

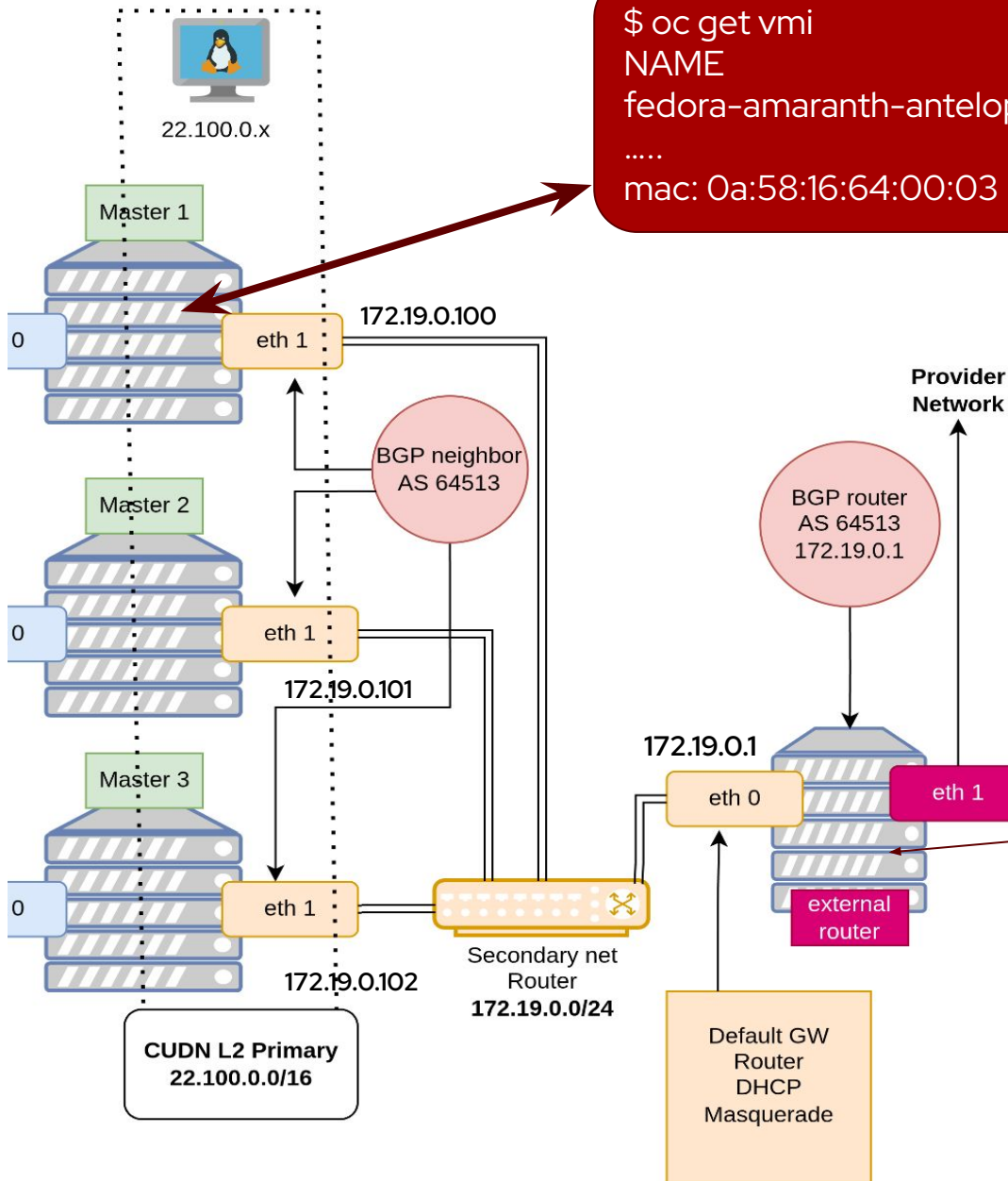
☐ Automatically selected Node

☒ Specific Node

	Name ↑	Status ↑	CPU ↑
☒	58-47-ca-7a-f6-73	✓ Ready	1.4 cores / 20 cores
☐	58-47-ca-7a-fb-7f	✓ Ready	1.5 cores / 20 cores

Migrate the VM to a different OCP
node





```
$ oc get vmi
NAME                                IP            NODENAME
fedora-amaranth-antelop2-2 22.100.0.3 58-47-ca-7a-f6-73
....
mac: 0a:58:16:64:00:03
```

FRR Router command -> "show evpn mac vni 100"

Number of MACs (local and remote) known for this VNI: 2

Flags: N=sync-neighs, I=local-inactive, P=peer-active, X=peer-proxy

MAC	Type	Flags	Intf/Remote	ES/VTEP	VLAN	Seq #'s
0a:58:16:64:00:03	remote		172.19.0.100			1/0
b6:bb:1c:86:82:77	local		br100		1	0/0

172.19.0.100 is the NEW VTEP of the OCP node where the VM is migrated



References

- [FRR Routing](#)
- [BGP RFC](#)
- [EVPN RFC](#)
- [OCP 4.22 CUDN BGP/EVPN](#)



Thank you



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